

Project 2 technical report:

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Dataset & Problem Framing

Dataset: Global Earthquake-Tsunami Risk Assessment Dataset — 782 complete records of global earthquakes with a magnitude of 6.5 or greater, recorded from 2001–2022. The dataset indicated no missing values, but just in case, any incomplete rows were dropped using `dropna()`.

Features:

Magnitude – moment magnitude (continuous 6.5 – 9.1)

Depth – the total depth in Km.

Latitude – geographical coordinate (continuous)

Longitude- geographical coordinate (continuous)

Target:

Tsunami – (1 = tsunami triggered, 0 = no tsunami).

Class distribution: 478 non-tsunami, 304 tsunami (61,39)

Research question: Given what we know about the factors of an earthquake like magnitude, depth, and location are we able to predict whether a tsunami will be triggered or not?

Relevance: Accurate assessment of tsunami risk from the variables of magnitude, depth, latitude, and longitude can support faster, more reliable alert systems.

Data Preprocessing & Assumptions

Feature selection: Under the name `core_features`, the main variables/features were isolated from a total of 12 columns. We are assuming that all 782 earthquakes in the dataset are accounted for.

```
core_features = ['magnitude', 'depth', 'latitude', 'longitude']
```

1. **Missing value handling:** Using `dropna()`, we remove any missing values that might be unaccounted for.

```
data = pd.concat([X, y], axis=1).dropna()
```

2. **Feature Scaling:** `StandardScaler` is applied to all features. This is required for logistic regression but is applied consistently across models for fairness.

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

3. **Train/test split:** we use a 80/20 stratified split and preserve the 61/39 ratio in both sets.

```
random_state=42,  
stratify=y
```

4. **Class imbalance Handling:** `class_weight='balanced'` is used in Random Forest to upweight the minority samples. Originally, it was used in logistic regression as well, but accuracy was taking a big hit, so I settled on not using it.

Key assumptions: All earthquakes have a magnitude of 6.5 or greater, meaning this model applies specifically to large earthquakes and should not be generalized to smaller earthquakes.

Model Selection & Experiments

A. Logistic Regression

- Config: `random_state=42, max_iter=5000`
- Rationale: Serves as a linear baseline for comparison, allowing us to determine whether a linear decision boundary is sufficient for classifying. `max_iter` was increased to 5000 to ensure the model fully converges.

B. Random Forest Classifier

- Config: `n_estimators=800, max_depth=None, random_state=42, class_weight='balanced'`
- Rationale: Random Forest is robust and handles outliers well. `n_estimators` is set to 800 to ensure stable predictions, and `class_weight='balanced'` is used to account for the 61/39 class imbalance between non-tsunami and tsunami events. `max_depth=None` allows the decision trees to grow without any depth limitations.

C. Gradient Boosting Classifier

- Config: `random_state=42, n_estimators=300, learning_rate=0.05`
- Rationale: Gradient Boosting builds trees one at a time, allowing errors from previous ones to be corrected. A lower `learning_rate` combined with `n_estimators=300` allows the model to learn more accurately while avoiding overfitting.

All Models that are used are held out on a 20% test set.

Evaluation & interpretation

Performance summary:

Model	Accuracy	AUC-ROC
Logistic Regression	58.6%	0.497
Random Forest	65.6%	0.739
Gradient Boosting	70.1%	0.769

Interpretation:

- Logistic Regression's AUC of 0.497 indicates that it is not a good fit in this feature space, as it performs near chance level.
- Random Forest improved its AUC to 0.730, demonstrating that the non-linear interactions between the geographic location features and depth are very important and significant.
- Gradient Boosting's AUC of 0.769 reflects its ability to focus on misclassified samples, showing its understanding of complex patterns that Random Forest misses.

Feature importance:

Feature	Gradient Boosting	Random Forest	Logistic Reg. (abs. coef.)
Longitude	0.597	0.400	0.376
Latitude	0.231	0.283	0.217
Depth	0.119	0.198	0.097
Magnitude	0.053	0.119	0.023

Geographical location (longitude and latitude combined) accounts for over 80% of Gradient Boosting's predictive power — geologically valid, considering most tsunamis occur over tectonic plates. Depth is ranked third, which is consistent with the shallow earthquakes in this dataset, and magnitude ranks last because of the range restriction, with all earthquakes being 6.5 or greater in this dataset.

Limitations, Ethics, Reflection

Limitations:

1. Sample Size: With our dataset only being 782 earthquakes, we could be missing or underrepresenting certain regions.
2. Limited Features: There are many other factors that contribute to earthquakes triggering tsunamis beyond the 4 features that we covered, and this could drastically change the results.
3. Class imbalance: The 61/39 split may be showing bias toward non-tsunami predictions, which could be impacting our results.
4. Scope: Our dataset only covers 2001–2022 and does not capture what earthquakes are like pre-2001 or post-2022. We are also only looking at earthquakes with magnitudes of 6.5 or greater, which limits our reach as well.

Ethical Considerations:

Deploying this model in a real-world sense, especially within real warning systems, carries serious responsibility. False negatives can cause missed tsunami predictions, which can lead to loss of life, while false positives can cause unnecessary evacuations and a loss of public trust. If this model were to be put into use, rigorous validation would need to be put into place, especially by experts. Also keep in mind that this is a prototype that could only be used for earthquakes with a magnitude of 6.5 or greater and should not be generalized for all earthquakes.

Reflection:

This project demonstrated that when it comes to predicting whether an earthquake will trigger a tsunami based on geographical location, depth, and magnitude, Gradient Boosting was the most consistent model. The dominance of geographical features over magnitude was certainly the most interesting finding — for large earthquakes, location matters more than how strong they are. For future work, I would work with a dataset that has a greater sample size and includes earthquakes with magnitudes below 6.5 as well.

Resources:

“Introduction to Global Historical Tsunami Data.” *National Centers for Environmental Information*, National Oceanic and Atmospheric Administration, <https://www.ncei.noaa.gov/products/natural-hazards/tsunamis-earthquakes-volcanoes/tsunamis/global-historical-data>.

Zaki, A. M. (n.d.). Global Earthquake–Tsunami Risk Assessment Dataset [Data set]. Kaggle. <https://www.kaggle.com/datasets/ahmeduzaki/global-earthquake-tsunami-risk-assessment-dataset>

AI Acknowledgement

Perplexity was used to troubleshoot errors and warning messages, and correct grammatical errors.